

Apply filters to SQL queries

Project description

Databases contain large amounts of data organized by tables. To access database data efficiently, search query language (SQL) is used to quickly search data and return information. Specific syntax is necessary to target data that is time consuming to find. Using SQL filter syntax such as WHERE, LIKE, OR, AND, and NOT helps searches to minimize the amount of data in the table and find what information is needed. In this project, I search through a database for specific information in a sample organization for employee and login information.

Retrieve after hours failed login attempts

My team discovered a security incident that occurred after business hours which can potentially be a brute force attempt. In this screenshot below, I use SQL filters to query the organization's database for failed login attempts. To search through all data I use **SELECT *** and FROM log_in_attempts to source the data from the specific information logs in the database. I use the **WHERE** argument to search for the after hour login time which is designated to be **18:00** or 6 o'clock. I use the **AND** to add an additional argument and use **= 0** to signal that logins were unsuccessful.

105	cjackson	2022-05-12	19:36:42	CAN	192.168.247.153	1
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
108	daquino	2022-05-09	21:30:48	CANADA	192.168.15.110	1
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
115	ivelasco	2022-05-10	23:06:01	CAN	192.168.154.1	1
116	tmitchel	2022-05-10	20:33:27	MEXICO	192.168.119.26	1
118	smartell	2022-05-12	23:21:31	MEXICO	192.168.173.196	1
119	tmitchel	2022-05-11	23:07:13	MEXICO	192.168.110.175	1
121	btang	2022-05-10	22:00:36	US	192.168.80.143	1
126	jrafael	2022-05-12	18:47:52	CAN	192.168.22.16	1
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
132	rjensen	2022-05-12	23:26:03	MEX	192.168.9.166	1
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
158	smartell	2022-05-09	19:30:32	MEXICO	192.168.190.178	1
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
164	jclark	2022-05-12	21:15:52	CAN	192.168.18.34	1
173	asundara	2022-05-12	23:17:52	US	192.168.58.217	1
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

39 rows in set (0.001 sec)

MariaDB [organization]> select * from log_in_attempts where login_time > '18:00' and success = 0;

event_id	username	login_date	login_time	country	ip address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

19 rows in set (0.001 sec)

MariaDB [organization]>

Retrieve login attempts on specific dates

A suspicious event occurred on the date 2022-05-09. To investigate the event, login attempts for the day of the event and the day before should be reviewed. In this screenshot below, I queried for login attempts on specific dates. Again I used **SELECT *** to access data categories and **FROM log_in_attempts** for login information. Then, I searched **WHERE login_date = '2022-05-09'** and the **OR** argument for **login_date '2022-05-08'**. This gave all data for the both dates inside of the SQL search while showing whether they were successful or not.

199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0
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19 rows in set (0.001 sec)

MariaDB [organization]> select *
 -> from log_in_attempts
 -> where login_date = '2022-05-09' or login_date = '2022-05-08';

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lynamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	aruso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
58	ivelasco	2022-05-09	17:20:54	CAN	192.168.57.162	0
61	dtanaka	2022-05-09	09:45:18	USA	192.168.98.221	1
65	aalonso	2022-05-09	23:42:12	MEX	192.168.52.37	1
66	aestrada	2022-05-08	21:58:32	MEX	192.168.67.223	1
67	abernard	2022-05-09	11:53:41	MEX	192.168.118.29	1
68	mrah	2022-05-08	17:16:13	US	192.168.42.248	1
70	tmitchel	2022-05-09	10:55:17	MEXICO	192.168.87.199	1
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
72	alevitsk	2022-05-08	12:09:10	CANADA	192.168.139.176	1
79	abernard	2022-05-09	11:41:15	MEX	192.168.158.170	0
80	cjackson	2022-05-08	02:18:10	CANADA	192.168.33.140	1
83	lrodriqu	2022-05-08	08:10:23	USA	192.168.67.69	1
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
90	gesparza	2022-05-09	00:49:05	CANADA	192.168.87.201	0
92	pwashing	2022-05-08	00:36:12	US	192.168.247.219	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
97	jreckley	2022-05-09	02:49:23	MEXICO	192.168.32.231	1
101	sbaelish	2022-05-08	12:01:22	US	192.168.145.158	0
102	jreckley	2022-05-09	16:51:44	MEX	192.168.108.13	1

Retrieve login attempts outside of Mexico

In this search, I searched for suspicious activity occurring outside of Mexico. To lookup this data I used **SELECT *** and **FROM log_in_attempts**. Then, I used the WHERE argument to search for countries that aren't Mexico. I added the argument **NOT country LIKE 'MEX%'**. The **NOT** argument negates the countries search which returns the opposite of the argument. The **LIKE** argument allows for searches that may contain the string starting with **'MEX'**. The **%** is positioned after the **MEX** in case that the country only uses the first 3 letters or the entire name depending on the spelling.

199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0
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56 rows in set (0.001 sec)

MariaDB [organization]> select *
 -> from log_in_attempts
 -> where not country like 'MEX%';

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrh	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:19:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
45	dcanaka	2022-05-11	10:28:54	US	192.168.223.157	1
46	eraab	2022-05-11	11:29:27	CAN	192.168.24.12	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
48	asundara	2022-05-11	03:18:45	USA	192.168.72.10	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
50	jclark	2022-05-10	10:48:02	CANADA	192.168.174.117	0
51	jrafael	2022-05-10	22:40:01	CANADA	192.168.148.115	1
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1

Retrieve employees in Marketing

My team wanted to perform security updates on specific employee machines in the Marketing department. I was responsible for getting information on these employee machines and will need to query the **employees** table. In this search, I used filters in SQL to create a query that identifies all employees in the Marketing department for all offices in the East building. This SQL filter required **SELECT *** and **FROM employees** to gather employee information. Then, I added an argument **WHERE department = 'Marketing'**. This argument searches for employees in the **Marketing** department. I added **AND** to join the two arguments together. Finally, the argument **office like 'EAST%'** searches for all offices in the east building. The % after **EAST** ensures that the search begins with east no matter the room number.

```
1063 | 1686ml40n569 | lpope | Sales | East-226 |
1069 | NULL | jpark | Finance | East-110 |
1072 | n905y920w694 | esmith | Sales | East-421 |
1074 | w622x645y348 | dcoleman | Information Technology | East-126 |
1075 | x573y883z772 | fbautist | Marketing | East-267 |
1077 | z654a154b259 | ldavis | Human Resources | East-241 |
1082 | e301f659g551 | mjohnson | Information Technology | East-151 |
1085 | h339i498j269 | cperez | Sales | East-325 |
1088 | k865l965m233 | rgosh | Marketing | East-157 |
1095 | r194s893t593 | glopez | Information Technology | East-457 |
1096 | s375t538u194 | kjeffers | Information Technology | East-419 |
1101 | x701y250z303 | ichowdhu | Human Resources | East-233 |
1103 | NULL | randers | Marketing | East-460 |
1109 | f229g533h679 | nlocklea | Sales | East-196 |
1113 | j781k420l510 | pjames | Human Resources | East-366 |
1121 | r628s557t397 | mrojas | Sales | East-288 |
1135 | f934g229h883 | khamamot | Information Technology | East-186 |
1140 | k982l199m839 | apatel2 | Human Resources | East-385 |
1142 | m674n127o823 | lsilva | Finance | East-440 |
1143 | n704o364p471 | sstark | Information Technology | East-282 |
1155 | z942a966b589 | zjones | Human Resources | East-122 |
1156 | a184b775c707 | dellery | Marketing | East-417 |
1159 | d881e710t732 | jshen | Finance | East-193 |
1161 | f951g408h866 | jjenkins | Information Technology | East-433 |
1163 | h679i515j339 | cwilliam | Marketing | East-216 |
1177 | v691w183x928 | aezra | Human Resources | East-190 |
1183 | b566c710d544 | lquraish | Human Resources | East-400 |
1187 | f963g637h851 | bbode | Finance | East-351 |
1192 | k570l183m949 | rlaghari | Information Technology | East-138 |
1195 | n516o853p957 | orainier | Finance | East-346 |
1199 | r520s571t459 | areyes | Human Resources | East-100 |
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41 rows in set (0.001 sec)

MariaDB [organization]> select *
-> from employees
-> where department = 'Marketing' and office like 'EAST%';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |
| 1103 | NULL | randers | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+
7 rows in set (0.033 sec)

MariaDB [organization]>
```

Retrieve employees in Finance or Sales

The security team required updates on the employee departments in the Sales and Finance departments. I inputted **SELECT *** to look through all information categories and **FROM employees** to retrieve employee information. Then, I added the argument **WHERE department = 'Sales'** to query all departments in Sales. The **OR** argument includes the 2nd part of the argument that queries the Finance department. I made sure to include **department = 'Finance'** as that includes both Sales and Finance departments.

Browser: SJU | URL: wi4uzccxk2s7shexjq.us-east1-c.resources.bumper-boats-00.services.qwiklabs.com | Tab: https://4ssuqr347nul...

```
MariaDB [organization]> select * from employees where department = 'Sales' or department = 'Finance';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h1741497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodrigu	Sales	South-134
1010	k2421212m542	jlansky	Finance	South-109
1011	l748m120n401	droas	Sales	South-292
1015	p611g262z945	jsoto	Finance	North-271
1017	r550a824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929g222z778	cgriffin	Sales	North-208
1044	s499c157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mscott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1069	NULL	jpark	Finance	East-110
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421
1076	y347z204a710	fgarcia	Finance	Central-270
1078	a667b270c984	sharley	Sales	North-418
1081	d647e310f618	qcorbit	Finance	South-290
1083	f840g812h544	gkoshi	Finance	West-165
1085	h339i498j269	cperez	Sales	East-325
1086	i281j129k749	lmajumda	Sales	West-499
1089	k38m929n154	jpark2	Sales	West-251
1091	n378o313p469	rtram	Sales	Central-230
1092	o391p779q935	lpark	Sales	West-227
1098	u671v146w618	tarchamb	Sales	North-423
1099	v283w690x104	anaser	Finance	West-357
1105	b551c837d758	kmei	Finance	Central-232
1107	d168e758f876	akajwara	Sales	North-471
1109	f229g533h679	nlocklea	Sales	East-196

Retrieve all employees not in IT

The security team required one more update for all departments excluding IT. IT had already received the update and all other departments required it. I created an SQL query beginning with **SELECT *** and **FROM employees** again. Then, I added 2 arguments **WHERE NOT** to lookup departments that are not Information Technology. The **NOT** negates the accepted argument which will return every other department.

33 rows in set, 5 warnings (0.001 sec)

```

MariaDB [organization]> select *
-> from employees
-> where not department = 'Information Technology';

```

employee_id	device_id	username	department	office
1000	a320b137c219	el Larson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f8775788	craab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i1497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodrigu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t220	jclark	Finance	North-188
1018	s310e540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrhah	Marketing	West-246
1028	c603d749e374	astrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	baand	Human Resources	East-484
1035	j236k303l245	bisales	Sales	South-171
1036	k550l533m205	rrjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260
1039	n253o917p623	cjackson	Sales	East-378
1040	o783p832q294	dtarly	Human Resources	East-237
1041	p929q222r778	cgriffin	Sales	North-208
1042	q175r338s833	acook	Human Resources	West-381
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1051	z451a308b518	itraora	Marketing	Central-134
1052	a192b174c940	jdarosa	Marketing	East-195
1053	b979c871d361	nemmanue	Human Resources	Central-259

Summary

Overall, this was a great introduction experience into SQL and database queries. Using different arguments to filter each query gave insight into how real-world database environments are utilized. The different arguments have purposes and garner information uniquely. This showed the significance of data efficiency as without adding filters would make data lookup time consuming.